

Spatial Non-stationary Near-Field Channel Estimation for mmWave Multi-User ELAA Systems via Low-Rank Tensor Decomposition

Jionghui Wang*, Jilin Wang*, Wenyan Ma[†]

*National Key Laboratory of Wireless Communications,

University of Electronic Science and Technology of China, Chengdu 611731, China

[†]Department of Electrical and Computer Engineering, National University of Singapore, Singapore 117583

Email: jionghuiwang@std.uestc.edu.cn, jilinwang@std.uestc.edu.cn, wenyan@u.nus.edu

Abstract—In this paper, we study the problem of uplink channel estimation for extremely large-scale millimeter-wave (mmWave) systems, where the base station (BS), equipped with a large-scale antenna array, serves multiple users in the same time-frequency resource block. Both near-field propagation and spatial non-stationarity are explicitly incorporated into the channel model. To efficiently estimate multi-user channels, we express the received signal as a third-order low-rank tensor that admits a canonical polyadic decomposition (CPD). An alternating least squares (ALS) algorithm is employed to obtain the factor matrices associated with the CPD, and channel parameters are efficiently estimated from the recovered factor matrices. The uniqueness analysis shows that the proposed method can work with non-orthogonal training sequences where the number of pilot symbols is much smaller than the number of users. This enables a substantial reduction in training overhead compared with conventional approaches that rely on orthogonal pilot sequences to decouple the multi-user channel estimation problem into independent single-user estimation tasks. Simulation results demonstrate that the proposed CPD-based method achieves a much higher estimation accuracy than compressed sensing-based methods.

Index Terms—Near-field channel estimation, millimeter-wave (mmWave) systems, spatial non-stationary (SnS), canonical polyadic decomposition (CPD).

I. INTRODUCTION

Millimeter-wave (mmWave) communications empowered by extremely large-scale antenna arrays (ELAAs) are considered a key enabling technology for the next-generation mobile communications to meet the ever-growing demand for higher data rates [1], [2]. The integration of ELAAs and the mmWave bands causes the propagation characteristics shift from far-field planar wavefronts to near-field spherical wavefronts [3], [4]. Besides, ELAA systems exhibit pronounced spatial non-stationarity (SnS), where different regions of the array may observe different scatterers/users [5]. These characteristics significantly increase the complexity of channel estimation in mmWave ELAA systems. In particular, near-field spherical

wavefronts introduce nonlinear phase variations with coupled angle–distance parameters, while SnS leads to region-dependent channel response across the array.

To model the near-field spherical wavefront, a polar-domain sparse representation is proposed in [6], where the near-field channel estimation to be formulated as a compressed sensing (CS) problem and solved via a variety of methods [6]–[9]. Further studies in [10], [11] show that near-field channels exhibit a block-sparse representation on a specially designed unitary matrix, yielding a well-conditioned measurement matrix that is more suitable for CS recovery. Beyond CS-based methods, tensor decomposition-based methods are developed for far-field mmWave/THz channel estimation [12] and later extended to near-field channel estimation [13], [14] by exploiting the inherent multi-dimensional channel structures. To address the SnS, a sub-array-based modeling framework is introduced in [15], which has motivated several sub-array-wise methods [16]–[18] for joint visibility region (VR) detection and near-field channel estimation.

Nevertheless, most existing works [6]–[10], [13], [16]–[18] rely on orthogonal pilot designs, which decompose multi-user estimation into independent single-user problems. For uplink transmission in ELAA systems, it is more attractive to consider joint multi-user channel estimation with non-orthogonal pilots, as this paradigm has the potential to support a growing number of users with limited pilot overhead.

In this work, by exploring the inherent multi-dimensional low-rank structure of the SnS near-field channels, we propose a CPD-based method for joint multi-user channel estimation. Specifically, the proposed method can guarantee to reliably estimate multiple users' channels even using non-orthogonal pilot sequences where the number of pilot symbols is much smaller than the number of users, thereby enabling joint multi-user channel estimation with a reduced training overhead. Simulation results also validate that the proposed method achieves superior estimation performance compared with CS-based approaches with the same number of pilot symbols.

II. SYSTEM MODEL

Consider the uplink channel estimation problem in an ELAA mmWave system, as shown in Fig. 1, where the base station (BS) is equipped with a large-scale uniform linear array (ULA) of N antennas and there are a total number of K single-antenna users that are served by the BS. The BS has $K \leq M \ll N$ RF chains. We assume the system has a bandwidth of B Hz and consists of $P \geq K$ sub-carriers. Let f_c denote the central carrier frequency, and the corresponding wavelength is $\lambda_c = c/f_c$, where c is the speed of light. Then, the frequency associated with the p -th sub-carrier is expressed as $f_p = f_c + \frac{2p-P}{2P}B$.

We focus on the line-of-sight (LoS)-dominated scenario, where there exists only a single LoS path between the BS and each user. Such an assumption is reasonable for mmWave communications, since the power of none-line-of-sight (NLoS) paths is typically much weaker, i.e., approximately 15 dB lower than that of the LoS path over the mmWave frequency bands [19].

In ELAA systems, the growing array aperture size gives rise to pronounced near-field effects as well as SnS. Specifically, the near-field effect requires adopting a more accurate spherical wavefront model to replace the conventional far-field wavefront model, while the SnS effect implies that a propagation path can only be observed by a subset of the array elements rather than the entire array, which is commonly referred to as the VR. Under these considerations, the channel between the k -th user and the BS at the p -th sub-carrier can be modeled as

$$\mathbf{h}_{p,k} = \alpha_k e^{-j2\pi f_p \tau_k} (\mathbf{b}(\theta_k, r_k) \otimes \mathbf{v}_k), \quad (1)$$

where $\{\alpha_k, \tau_k, \theta_k, r_k\}$ denote the complex channel gain, the time delay, the angle and the distance from the reference antenna to the k -th user, respectively. The operator $\otimes$ denotes the Hadamard product. $\mathbf{b}(\theta_k, r_k)$ represents the near-field steering vector, which is given by

$$\mathbf{b}(\theta_k, r_k) = \frac{1}{\sqrt{N}} [e^{-j\frac{2\pi}{\lambda_c}(r_k^{(1)} - r_k)} \dots e^{-j\frac{2\pi}{\lambda_c}(r_k^{(N)} - r_k)}]^T, \quad (2)$$

where $r_k^{(n)}$ denotes the distance between the k -th user and the n -th antenna of the BS. For simplicity, we choose the first antenna as the reference antenna and we have $r_k^{(1)} \equiv r_k$. According to the geometric relationship, $r_k^{(n)}$ is given by $r_k^{(n)} = \sqrt{r_k^2 + (n-1)^2 d^2 - 2r_k(n-1)d \sin \theta_k}$. Note that $\mathbf{v}_k$ is a binary vector whose n -th element equals 1 if $n \in \mathcal{V}_k$, and 0 otherwise, where $\mathcal{V}_k$ denotes the set of array element indices that can observe the path of the k -th user.

To further reduce the estimation complexity, we exploit the fact that antennas within a local region of the array can be assumed to share the same visibility status. Specifically, the array is partitioned into N_s sub-arrays with each consisting of $U = N/N_s$ antennas. Under this partition, the original visibility vector $\mathbf{v}_k$ can be reduced to a lower-dimensional binary vector $\tilde{\mathbf{v}}_k \in \mathbb{B}^{N_s}$, whose n_s -th element characterizes the visibility status of the n_s -th sub-array. The two vectors are

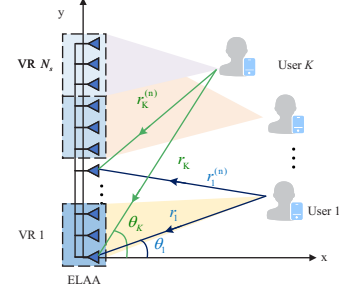


Fig. 1. Multi-user SnS Near-field ELAA systems.

related by $\mathbf{v}_k = \tilde{\mathbf{v}}_k \otimes \mathbf{1}_U$, where $\mathbf{1}_U \in \mathbb{R}^U$ denotes an all-ones vector. In general, a larger N_s leads to a finer visibility resolution and hence a more accurate estimate [16].

III. PROBLEM FORMULATION

A. Received Signal Model

The training sequence consists of T consecutive orthogonal frequency division multiplexing (OFDM) symbols. At the t -th time instant, each user sends its pilot vector $\mathbf{s}_k(t) = [s_{k,1}(t) \ s_{k,2}(t) \ \dots \ s_{k,P}(t)]^T \in \mathbb{C}^P, \forall k, t$ to the BS, where $s_{k,p}(t)$ denotes the t -th pilot symbol associated with the k -th user and the p -th sub-carrier. At the BS, each RF chain, say the m -th RF chain, employs an individual combining vector $\mathbf{w}_m \in \mathbb{C}^N$ to combine the received antenna signals. Then, the cyclic prefix is removed and symbols are converted back to the frequency domain by performing a discrete Fourier transform (DFT). After processing, the received signal associated with the p -th sub-carrier can be expressed as

$$y_{p,m}(t) = \mathbf{w}_m^H \sum_{k=1}^K \mathbf{h}_{p,k} s_{k,p}(t) + \mathbf{w}_m^H \mathbf{n}_p(t), \quad (3)$$

where $\mathbf{n}_p(t) \in \mathbb{C}^N$ is the additive white Gaussian noise (AWGN).

Collecting the received signals from all M RF chains yields

$$\mathbf{y}_p(t) = \mathbf{W}^H \sum_{k=1}^K \mathbf{h}_{p,k} s_{k,p}(t) + \tilde{\mathbf{n}}_p(t), \quad (4)$$

where $\mathbf{y}_p(t) \triangleq [y_{p,1}(t) \ \dots \ y_{p,M}(t)]^T \in \mathbb{C}^M$; $\mathbf{W} \triangleq [\mathbf{w}_1 \ \dots \ \mathbf{w}_M] \in \mathbb{C}^{N \times M}$ denotes the combining matrix; and $\tilde{\mathbf{n}}_p(t) \triangleq \mathbf{W}^H \mathbf{n}_p(t) \in \mathbb{C}^M$ is the equivalent Gaussian noise after combining.

Suppose that pilot symbols are identical across all sub-carriers, i.e. $s_{k,p}(t) = s_k(t), \forall p$ and define $\mathbf{s}_k \triangleq [s_k(1) \ \dots \ s_k(T)]^T \in \mathbb{C}^T$. Then, the received signal at the p -th sub-carrier can be written as

$$\mathbf{Y}_p = \sum_{k=1}^K \alpha_k e^{-j2\pi f_p \tau_k} \mathbf{c}_k(\alpha_k, \theta_k, r_k, \mathbf{v}_k) \mathbf{s}_k^T + \tilde{\mathbf{N}}_p, \quad (5)$$

where $\mathbf{Y}_p \triangleq [\mathbf{y}_p(1) \ \dots \ \mathbf{y}_p(T)] \in \mathbb{C}^{M \times T}$, $\mathbf{c}_k(\alpha_k, \theta_k, r_k, \mathbf{v}_k) \triangleq \alpha_k \mathbf{W}^H (\mathbf{b}(\theta_k, r_k) \otimes \mathbf{v}_k)$ is the compressed spatial array response vector, which jointly captures the complex path gain, the near-field spherical wavefront and the SnS effect associated with the k -th user, and $\tilde{\mathbf{N}}_p \triangleq [\tilde{\mathbf{n}}_p(1) \ \dots \ \tilde{\mathbf{n}}_p(T)] \in \mathbb{C}^{M \times T}$;

B. Low-Rank Tensor Representation

By putting all subcarriers' signals $\{\mathbf{Y}_p\}_{p=1}^P$ together, we can construct a third-order tensor $\mathcal{Y} \in \mathbb{C}^{P \times M \times T}$, whose three modes respectively represent the subcarrier, the RF chain and the OFDM symbol. It can be readily verified that the tensor $\mathcal{Y}$ admits a canonical polyadic decomposition (CPD), i.e., it can be expressed as a sum of rank-one component tensors:

$$\mathcal{Y} = \sum_{k=1}^K \mathbf{g}(\tau_k) \circ \mathbf{c}_k(\alpha_k, \theta_k, r_k, \mathbf{v}_k) \circ \mathbf{s}_k + \mathcal{N}, \quad (6)$$

where $\circ$ denotes the outer product operation, $\mathcal{N} \in \mathbb{C}^{M \times T \times P}$ is the equivalent noise tensor, and $\mathbf{g}(\tau_k) \triangleq [e^{-j2\pi f_1 \tau_k} \dots e^{-j2\pi f_P \tau_k}]^T \in \mathbb{C}^P$ denotes the frequency response across all subcarriers. Furthermore, define

$$\begin{aligned} \mathbf{G} &\triangleq [\mathbf{g}(\tau_1) \dots \mathbf{g}(\tau_K)] \in \mathbb{C}^{P \times K}, \\ \mathbf{C} &\triangleq [\mathbf{c}_1(\alpha_1, \theta_1, r_1, \mathbf{v}_1) \dots \mathbf{c}_K(\alpha_K, \theta_K, r_K, \mathbf{v}_K)] \in \mathbb{C}^{M \times K}, \\ \mathbf{S} &\triangleq [\mathbf{s}_1 \dots \mathbf{s}_K] \in \mathbb{C}^{T \times K}, \end{aligned} \quad (7)$$

which are the factor matrices of a noiseless version of $\mathcal{Y}$.

The CPD form (6) of the received signals allows us to find a more efficient way to estimate the wireless channel. Specifically, under some mild conditions that will be discussed later, the CPD is unique up to scaling and permutation ambiguities. Thus we can first estimate the three factor matrices and then extract the associated channel parameters $\{\alpha_k, \tau_k, \theta_k, r_k, \mathbf{v}_k\}$ from the recovered factor matrices.

IV. CPD-BASED CHANNEL ESTIMATION

A. Uniqueness Condition

Before performing CPD, we briefly discuss its uniqueness condition, which is essential for reliable channel parameter estimation. Let k_A denote the k -rank of $\mathbf{A}$, which is defined as the maximum number k_A such that any k_A columns of $\mathbf{A}$ are linearly independent. Based on Kruskal's condition [20], the CPD of the third-order tensor $\mathcal{Y}$ in (6) is essentially unique (up to scaling and permutation ambiguities), provided that the following condition holds:

$$k_G + k_C + k_S \geq 2K + 2. \quad (8)$$

We first examine the k -rank of $\mathbf{G}$. Since $\mathbf{G}$ is a column-wise scaled Vandermonde matrix characterized by distinct delay parameters $\{\tau_k\}$, the k -rank of $\mathbf{G}$ is equal to the smallest value of P and K , i.e., $k_G = \min(P, K) = K$.

We now examine the k -rank of $\mathbf{C}$. Recall that the factor matrix $\mathbf{C}$ can be expressed as $\mathbf{C} = \mathbf{W}^H \mathbf{D}_\alpha \mathbf{B}_v$ where $\mathbf{D}_\alpha \triangleq \text{diag}(\alpha_1, \dots, \alpha_K)$ and $\mathbf{B}_v \triangleq [\mathbf{b}(\theta_1, r_1) \otimes \mathbf{v}_1 \dots \mathbf{b}(\theta_K, r_K) \otimes \mathbf{v}_K]$. In general, the columns of $\mathbf{B}_v$ correspond to distinct parameters and are therefore linearly independent, which implies $\mathbf{B}_v$ has full column rank. When a random analog beamforming matrix $\mathbf{W}$ is adopted, it follows that the k -rank of $\mathbf{C}$ is $k_C = \min(M, K) = K$ with probability one [12].

By taking the above results back to (8), we can conclude that the Kruskal condition is satisfied as long as $k_S \geq 2$, i.e., any two columns of $\mathbf{S}$ are linearly independent. This condition can be readily fulfilled provided that $T \geq 2$ and the pilot sequences

are linearly independent. In practice, such training sequences are often obtained by minimizing the mutual coherence among the columns $\mathbf{S}$, which has been extensively studied under the framework of the grassmannian line packing problem [21]–[23].

From the above analysis, we know that the Kruskal's condition can be met even when $T \ll K$ (provided that $T \geq 2$). As a result, all users' channels can be jointly estimated without requiring $T \geq K$ pilot symbols. This enables a substantial reduction in the training overhead as we only need to use a minimum of $T = 2$ OFDM pilot symbols to complete the multi-user channel estimation task.

B. CPD

The CPD can be accomplished by solving the following optimization problem

$$\min_{\hat{\mathbf{G}}, \hat{\mathbf{C}}, \hat{\mathbf{S}}} \|\mathcal{Y} - \sum_{k=1}^K \hat{\mathbf{g}}_k \circ \hat{\mathbf{c}}_k \circ \hat{\mathbf{s}}_k\|_F^2. \quad (9)$$

Here the number of users K is known *a priori*, since it is determined by the user scheduling and access control procedures at the BS, which are performed at a higher protocol layer before channel estimation and signal processing. $\hat{\mathbf{g}}_k$, $\hat{\mathbf{c}}_k$ and $\hat{\mathbf{s}}_k$ denote the k -th column of $\hat{\mathbf{G}}$, $\hat{\mathbf{C}}$ and $\hat{\mathbf{S}}$, respectively. The above optimization can be efficiently solved by an alternating least squares (ALS) procedure [20], which iteratively minimizes the data fitting error with respect to the three factor matrices.

Based on the uniqueness analysis presented earlier, the CPD in (6) is unique up to scaling and permutation ambiguities. Accordingly, ignoring the estimation errors for simplicity, the estimated factor matrices are related to the true ones as $\hat{\mathbf{G}} = \mathbf{G}\mathbf{\Lambda}_1\mathbf{\Pi}$, $\hat{\mathbf{C}} = \mathbf{C}\mathbf{\Lambda}_2\mathbf{\Pi}$ and $\hat{\mathbf{S}} = \mathbf{S}\mathbf{\Lambda}_3\mathbf{\Pi}$, where $\mathbf{\Lambda}_1$, $\mathbf{\Lambda}_2$ and $\mathbf{\Lambda}_3$ are non-singular diagonal matrices satisfying $\mathbf{\Lambda}_1\mathbf{\Lambda}_2\mathbf{\Lambda}_3 = \mathbf{I}$, and $\mathbf{\Pi}$ is a permutation matrix.

C. Channel Estimation

We now discuss how to extract the channel parameters from the estimated factor matrices. It can be observed that each column of $\hat{\mathbf{S}}$ is a scaled version of the pilot sequence associated with a certain user. Note that all users' training sequences are known *a priori* at the BS. Hence, the permutation matrix $\mathbf{\Pi}$ can be estimated using a simple correlation-based method. Once $\mathbf{\Pi}$ is obtained, the permutation ambiguity for the estimated factor matrices can be removed, and thus we have $\hat{\mathbf{G}} = \mathbf{G}\mathbf{\Lambda}_1$, $\hat{\mathbf{C}} = \mathbf{C}\mathbf{\Lambda}_2$ and $\hat{\mathbf{S}} = \mathbf{S}\mathbf{\Lambda}_3$.

It can be observed that the k -th column of $\hat{\mathbf{G}}$ and $\hat{\mathbf{C}}$ are the scaled version of $\mathbf{g}(\tau_k)$ and $\mathbf{c}_k(\alpha_k, \theta_k, r_k, \mathbf{v}_k)$, respectively. Consequently, the time delay τ_k can be estimated by

$$\hat{\tau}_k = \arg \max_{\tau_k} \frac{|\mathbf{g}^H(\tau_k) \hat{\mathbf{g}}_k|}{\|\mathbf{g}(\tau_k)\|_2 \|\hat{\mathbf{g}}_k\|_2}, \quad (10)$$

which can be efficiently solved via a one-dimensional line search. Moreover, for the LoS scenario, the distance from the BS to the k -th user can be readily obtained as $\hat{r}_k = \hat{\tau}_k c$.

Since $\hat{r}_k$ has already been obtained, we only need to estimate θ_k and $\mathbf{v}_k$, which can be formulated as the following problem:

$$\min_{\mathbf{v}_k, \theta_k} \|\hat{\mathbf{c}}_k - \lambda_{2,k} \mathbf{W}^H (\mathbf{b}(\theta_k, \hat{r}_k) * \mathbf{v}_k)\|_2^2, \quad (11)$$

where $\lambda_{2,k}$ denotes the k -th diagonal element of $\mathbf{\Lambda}_2$. Fixing $\mathbf{v}_k$ and θ_k , the optimal scaling coefficient $\lambda_{2,k}$ can be obtained in closed form by setting the gradient of the objective function to zeros, i.e.,

$$\lambda_{2,k}^* = \frac{(\mathbf{b}(\theta_k, \hat{r}_k) \otimes \mathbf{v}_k)^H \mathbf{W} \hat{\mathbf{c}}_k}{\|\mathbf{W}^H (\mathbf{b}(\theta_k, \hat{r}_k) \otimes \mathbf{v}_k)\|_2^2}. \quad (12)$$

Substituting (12) into (11), the original problem can be simplified as a normalized correlation maximization problem given by

$$\max_{\mathbf{v}_k, \theta_k} \frac{|\hat{\mathbf{c}}_k^H \mathbf{W}^H (\mathbf{b}(\theta_k, \hat{r}_k) \otimes \mathbf{v}_k)|}{\|\hat{\mathbf{c}}_k\|_2 \|\mathbf{W}^H (\mathbf{b}(\theta_k, \hat{r}_k) \otimes \mathbf{v}_k)\|_2}, \quad (13)$$

which can be efficiently solved via alternating optimization. Specifically, for a fixed $\mathbf{v}_k$, θ_k is updated by one-dimensional line search; whereas for a fixed θ_k , $\mathbf{v}_k$ (or equivalently its reduced-dimensional representation $\tilde{\mathbf{v}}_k$) is updated via a brute-force search.

Next, we proceed to estimate the path loss $\{\hat{\alpha}_k\}$. After obtaining the estimates $\{\hat{\theta}_k, \hat{r}_k, \mathbf{v}_k\}$, the factor matrices $\tilde{\mathbf{C}}$ can be reconstructed as $\tilde{\mathbf{C}} = \mathbf{W}^H [\mathbf{b}(\hat{\theta}_1, \hat{r}_1) \otimes \hat{\mathbf{v}}_1 \cdots \mathbf{b}(\hat{\theta}_K, \hat{r}_K) \otimes \hat{\mathbf{v}}_K]$, and the scaling matrix $\mathbf{\Lambda}_2$ can be obtained according to (12). Ideally, there should be $\tilde{\mathbf{C}} = \tilde{\mathbf{C}} \mathbf{D}_\alpha \mathbf{\Lambda}_2$. Accordingly, $\mathbf{D}_\alpha$ can be estimated as $\hat{\mathbf{D}}_\alpha = \tilde{\mathbf{C}}^\dagger \tilde{\mathbf{C}} \mathbf{\Lambda}_2^{-1}$. Finally, the channels $\{\mathbf{h}_{p,k}\}$ can be reconstructed according to (1).

V. SIMULATION RESULTS

We provide simulation results to evaluate the performance of the proposed CPD-based joint multi-user channel estimation method for SnS near-field ELAA systems. We consider a BS equipped with a ULA with $N = 128$ antennas and $M = 32$ RF chains, and the carrier frequency is set to $f_c = 30$ GHz with the bandwidth of $B = 0.2$ GHz and the number of subcarriers of $P = 64$. Accordingly, the Rayleigh distance can be computed as $d_R = 80.6$ m. The entire array is partitioned into $N_s = 4$ sub-arrays, each consisting $U = 32$ antennas. $K = 8$ users are assumed to be located within the near-field region of the BS. For each user, the angle θ_k and the distance r_k from the reference antenna to each user are randomly chosen from $[20 \text{ m}, 80 \text{ m}]$ and $[-60^\circ, 60^\circ]$, respectively. The complex path gain α_k is generated according to [24]. The pilot symbol matrix $\mathbf{S}$ is generated following [22] for $T = 2$, [23] for $3 \leq T < 8$, and is chosen as a DFT matrix when $T = 8$. The signal-to-noise-ratio (SNR) is defined as $\text{SNR} \triangleq \|\mathcal{Y} - \mathcal{N}\|_F^2 / \|\mathcal{N}\|_F^2$.

Furthermore, the proposed CPD-based is compared with the group time block code (GTBC)-based polar-domain simultaneous orthogonal matching pursuit (GP-SOMP) algorithm presented in [16]. Since [16] considers only orthogonal pilots, we extend this method to the non-orthogonal pilot scheme by

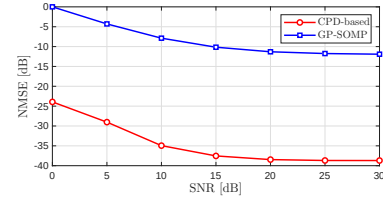


Fig. 2. NMSE versus SNR ($T = 4$)

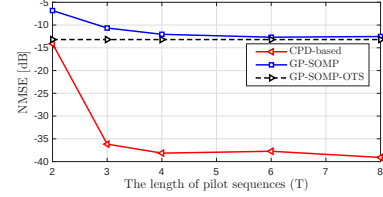


Fig. 3. NMSE versus the length of pilot sequences (SNR = 30 dB)

modifying the CS dictionary from $\mathbf{W}_U$ to $\mathbf{S} \otimes \mathbf{W}_U$ to ensure a fair comparison. The normalized mean square error (NMSE) is adopted as the performance metric, which is calculated as $\text{NMSE} \triangleq \frac{\sum_{p=1}^P \sum_{k=1}^K \|\mathbf{h}_{p,k} - \hat{\mathbf{h}}_{p,k}\|_F^2}{\sum_{p=1}^P \sum_{k=1}^K \|\mathbf{h}_{p,k}\|_F^2}$.

Fig. 2 plots the NMSE performance of respective algorithms versus SNR (dB), where we set $T = 4$. It can be observed that the proposed CPD-based method achieves a significant performance improvement over the GP-SOMP method when non-orthogonal training sequences are employed. This is attributed to the fact that the proposed method effectively explores the intrinsic multi-dimensional low-rank structure of the SnS near-field channels.

In Fig 3, we evaluate the NMSE performance versus the pilot length T , where the SNR is set to 30 dB and the term “GP-SOMP-OTS” denotes the conventional scheme which converts multi-user channel estimation into independent single-user channel estimation problems by using orthogonal training sequences [16]. We see that when $T \geq 2$, the proposed CPD-based method can achieve a reliable channel estimate, which coincides well with the aforementioned uniqueness analysis. Also, the proposed CPD-based method shows a significant performance advantage over the GP-SOMP-OTS scheme in terms of both estimation accuracy and training overhead expenses. This is because the CPD can naturally decorrelate the users’ channels and enables a unique factorization of the received signal into distinct components associated with time delays, angle-distance pairs and VR status. As a result, the channel parameters of each user can be more accurately estimated.

VI. CONCLUSIONS

In this paper, we developed a CPD-based SnS near-field channel estimation method for mmWave MU-MIMO-OFDM systems. The proposed method can effectively explore the intrinsic multi-dimensional low-rank structure of the channels, allowing an additional pilot overhead reduction. Simulation results show that the proposed method exhibits a significant performance improvement over the CS-based algorithm.

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